COMP1870 Theoretical Foundations of Computer Science I

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Solutions to worksheet for week 1.3

1. Let a and b be positive integers. We want to prove that if a and b are odd, then $a + b$ is even.

   (a) Is there a best way to prove this? Direct proof, contrapositive proof, or proof by contradiction?
   Solution. My feeling is that the simplest way to prove this is via a direct proof. You might ask how you are supposed to know this, and this is a legitimate question. The truthful answer is that after working in mathematics for a while, you develop an intuition for the right approach to a proof. It takes time and practice to obtain this intuition — don't feel bad if you don't immediately have a sense of how to start a proof. The point of this exercise is to start building this intuition.

   (b) Can you give a proof (using some method)?
   Solution. What does it mean when we say that a positive integer is even? We mean that this number can be expressed as the product of two and an integer. So a being even means that a can be written as $2m$, where m is some integer. Saying that a number is odd is the same as saying that it is one more than an even number. So if a is odd means that we can write it as $2m+1$ where m is an integer. Similarly, if b is odd, then it can be written as $2n+1$, where n is some integer. (Note that we have used a new variable n, rather than reusing m, since we don't know whether or not m and n are equal.) Now we try calculation.

   $$a + b = (2m + 1) + (2n + 1) = (2m + 2n) + (1 + 1) = 2(m + n) + 2 = 2(m + n + 1).$$

   This means that $a+b$ can be expressed as the product of two and an integer, in particular the integer $m + n + 1$. This means that $a + b$ is even. We have reached our conclusion, so the proof is complete.

2. Let a, b, and p be positive integers, where p is prime. Imagine that we want to prove that if p divides ab, then either p divides a or p divides b.

   (a) If you wrote a direct proof, how would you start the proof? How would you end it?
   Solution. Assume that p divides ab. ... Therefore either p divides a or p divides b.

   (b) If you wrote a contrapositive proof, how would you start the proof? How would you end it?
   Solution. Assume that p does not divide a and p does not divide b. ... Therefore p does not divide ab.

   (c) If you wrote a proof by contradiction, how would you start the proof? How would you end it?
   Assume that p divides ab, and that p does not divide a or b. ... This is impossible, so we have reached a contradiction.

3. Let $x_1, x_2, \ldots, x_n$ be a sequence of positive integers. We want to prove that if $x_1 + x_2 + \cdots + x_n = k$, then at least one of the numbers in $x_1, x_2, \ldots, x_n$ is greater than or equal to k/n.

   (a) Is there a best way to prove this? Direct proof, contrapositive proof, or proof by contradiction?
   Solution. My instinct is that a proof by contradiction will give the simplest proof, although other strategies are certainly possible.

   (b) Can you give a proof (using some method)?
   Solution. Assume for a contradiction the conclusion is false. The conclusion is an existential statement: there exists at least one number that is greater than or equal to k/n. The negation of this statement is universal: every number x_i is less than k/n. We assume this negation is true, and try to derive a contradiction. Since each x_i is less than k/n, the sum $x_1+x_2+\cdots+x_n$ is less than $n \times (k/n) = k$. This contradicts the assumption that the sum is equal to k. We have reached an impossible contradiction so the proof is complete.

4. We want to write a proof of the following fact: if n is a positive integer and n^2 is odd, then n is odd.

   (a) Construct a proof by completing the following statements.

   (1) Assume that n is even.

   (2) Because n is even, it can be expressed as $2k$, where k is some integer

   (3) Therefore n^2 is equal to $(2k)^2 = 4k^2 = 2(2k^2)$.

   (4) Because n^2 can be expressed as the product of two and an integer, it follows that n^2 is even.

   (b) What style of proof have we just used to show that if n^2 is odd then n is odd?

   Solution. We have proved that if n is even then n^2 is even. Equivaently, we proved that if *not odd* then n^2 is *not odd*. This is the contrapositive of the statement: if n^2 is even then n is even, so our proof is a contrapositive proof.

5. Give an inductive proof that

$$1 + 2 + 4 + \cdots + 2^n = 2^{n+1} - 1$$

   for every positive integer n. Make sure you check the base case, state the inductive hypothesis, and explain when you are using the inductive hypothesis.

   Solution. The base case is when n is the smallest positive integer, 1. When $n = 1$, the left-hand side of the equation is $1 + 2^1 = 1 + 2 = 3$. The right-hand side is $2^{1+1} - 1 = 2^2 - 1 = 4 = 1 = 3$. So the statement holds when $n = 1$, and therefore we have checked the base case.

   Now we make the inductive assumption that the statement is true when $n = k$. We will use this assumption to prove that the statement is true when $n = k + 1$. Note that

$$1 + 2 + 4 + \cdots + 2^k + 2^{k+1} = [1 + 2 + 4 + \cdots + 2^k] + 2^{k+1} = [2^{k+1} - 1] + 2^{k+1}$$

   where we have replaced the sum in the square brackets by using the inductive assumption. Now

$$[2^{k+1} - 1] + 2^{k+1} = 2 \cdot 2^{k+1} - 1 = 2^{k+2} - 1$$

   so we see that

$$1 + 2 + 4 + \cdots + 2^k + 2^{k+1} = 2^{k+2} - 1$$

   which means that the statement is true when $n = k + 1$. This completes the proof by induction.

6. Give an inductive proof that

$$2 + 4 + 6 + \cdots + 2n = n^2 + n$$

   for every positive integer n. Make sure you check the base case, state the inductive hypothesis, and explain when you are using the inductive hypothesis.

   Solution. The base case is when $n = 1$. In this case, the left-hand side of the equation is 2. The right-hand side is $1^2 + 1 = 2$, so the equation is true when $n = 1$. Now we make the inductive assumption and we assume that the equation is true when $n = k$. We will use this assumption to prove that the equation is true for $n = k + 1$. Note that

$$2 + 4 + 6 + \cdots + 2k + 2(k + 1) = [2 + 4 + 6 + \cdots + 2k] + 2(k + 1) = [k^2 + k] + 2(k + 1)$$

   where we have used the inductive assumption to replace the sum in the square brackets. We continuing calculating and we see that

$$(k^2 + k) + 2(k + 1) = k^2 + k + 2k + 2 = k^2 + 3k + 2 = (k^2 + 2k + 1) + (k + 1) = (k + 1)^2 + (k + 1).$$

Therefore

$$2 + 4 + 6 + \cdots + 2k + 2(k+1) = (k+1)^2 + (k+1)$$

so the equation is true when $n = k+1$. This completes the proof by induction.

7. Remember that a number p is divisible by 8 if it can be expressed as $p = 8q$, where q is some integer.

   Use induction on n to prove that $(2n+1)^2 - 1$ is divisible by 8 for every positive integer n. Make sure you check the base case, state the inductive hypothesis, and explain when you are using the inductive hypothesis.

   Solution.

   The base case is when $n = 1$. In this case, $(2n+1)^2 - 1$ is equal to $(2 \cdot 1 + 1)^2 - 1 = 3^2 - 1 = 9 - 1 = 8 = 8 \cdot 1$. This shows that $(2n+1)^2 - 1$ is divisible by 8 when $n - 1$. Therefore the base case holds.

   Now we make the inductive assumption that $(2k+1)^2 - 1$ is divisible by 8, for some positive integer k. We will use this assumption to prove that $(2(k+1)+1)^2 - 1$ is divisible by 8 and this will complete the inductive proof. Since $(2k+1)^2 - 1$ is divisible by 8, we express it as $8q$, for some integer q. Therefore

$$8q = (2k+1)^2 - 1 = (4k^2 + 4k + 1) - 1 = 4k^2 + 4k.$$

   Now we calculate

$$(2(k+1)+1)^2 - 1 = (2k+2+1)^2 - 1 = (2k+3)^2 - 1 = (4k^2+12k+9) - 1 = 4k^2+4k+8k+8 = 8q+8(k+1) = 8(q+k+1).$$

   We used the inductive hypothesis when we replaced $4k^2 + 4k$ with $8q$, since $4k^2 + 4k = 8q$ is equivalent to $(2k+1)^2 - 1 = 8q$. Since $(2(k+1)+1)^2 - 1$ can be expressed as the product of 8 and the integer $q + k + 1$, it is divisible by 8, which is what we wanted to show.

8. Let n be an integer satisfying $n \geq 3$. Use induction to prove that a polygon with n corners can be covered by $n - 2$ triangles in such a way that the triangles do not overlap, and the corners of each triangle line up with corners of the original polygon. (Assume that the polygon is *convex*, meaning that every corner has an internal angle less than 180°.)

   Solution. We prove this by induction on n. The base case is when $n = 3$. In this case, the polygon is a triangle, and any triangle can be divided into a single triangle, in a trivial way. So the base case holds.

   Now we make the inductive assumption. We assume that the statement is true when $n = k$. We will use this assumption to prove that the statement is true when $n = k+1$. Let P be a polygon with $k+1$ corners. Select a corner C of the polygon. Let C_1 and C_2 be the two neighbouring corners of C. We draw a line between C_1 and C_2 to create a triangle with C, C_1, and C_2 as its corners. Because the polygon is convex, this triangle is contained inside the polygon. We slice this triangle off P, and this produces a polygon with k corners. By the inductive assumption, this polygon can be divided into $k - 2$ triangles. When we add the triangle with C, C_1, and C_2 as corners, it follows that P can be divided into $k - 1 = (k+1) - 2$ triangles, so the proof is complete.

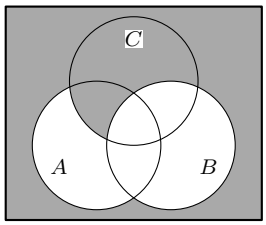
10. Let A, B, and C be sets. Draw Venn diagrams that represent each of the following sets. (Remember that you need to include the universal set in your Venn diagrams.)

    (a) $\overline{(A \cup B) \cap (B \cup \overline{C})}$

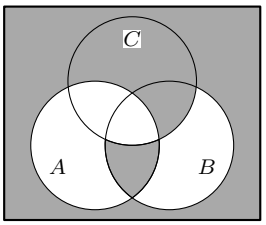
    (b) $A \triangle (\overline{B \cap \overline{C}})$

    (c) $(A \cap B) \setminus \overline{C}$

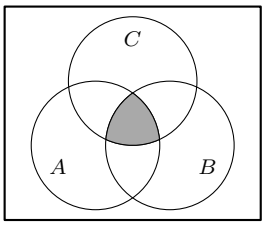
Solution. The sets are indicated by shaded areas in the following Venn diagrams.



(a)



(b)



(c)

11. Using the set identities in Proposition 5.12 of the lecture notes, prove that $\overline{\overline{A} \cup (B \cap \overline{A})}$ is equal to $A \cap (A \cup \overline{B})$. Explicitly state which identity you are using at each step.

    Why is $A \cap (A \cup \overline{B})$ equal to A?

    Solution.

$$\begin{array}{rcll} \overline{\overline{A} \cup (B \cap \overline{A})} & = & \overline{\overline{A}} \cap (\overline{B \cap \overline{A}}) & \text{De Morgan's law} \\ & = & A \cap (\overline{B \cap \overline{A}}) & \text{Double complements} \\ & = & A \cap (\overline{B} \cup \overline{\overline{A}}) & \text{De Morgan's law} \\ & = & A \cap (\overline{B} \cup A) & \text{Double complements} \\ & = & A \cap (A \cup \overline{B}) & \text{Commutative law} \end{array}$$

    Let X and Y be sets and assume that X is a subset of Y. We will prove that $X \cap Y = X$. Every element of $X \cap Y$ is in X so $X \cap Y \subseteq X$. Also, every element of X is in Y because $X \subseteq Y$, and therefore every element of X is in $X \cap Y$. Every element of X is in $X \cap Y$ and every element of $X \cap Y$ is in X, so the two sets are the same.

    Now every element of A is in $A \cup \overline{B}$, so A is a subset of $A \cup \overline{B}$. Therefore $A \cap (A \cup \overline{B})$ is equal to A, by the fact we have just proved.

12. Assume that U is the universal set. Why is $B \cup \overline{B} = U$ true? Why is $A \cap U = A$ true?

    Using these facts and the identities in Proposition 5.12, prove that $(A \cap B) \cup (A \cap \overline{B}) = A$.

    Solution. It is definitely true that every element of $B \cup \overline{B}$ is in U, because U is the universal set. On the other hand, let x be an arbitrary element of U. If x is in B, then x is in $B \cup \overline{B}$. If x is not in B, then it is in $\overline{B}$, so again it is in $B \cup \overline{B}$. So in either case x is in $B \cup \overline{B}$. This shows that every element of U is in $B \cup \overline{B}$. Since these two sets have the same elements, we have proved that $U = B \cup \overline{B}$.

    For the other fact, every element of $A \cap U$ is in A, by the definition of intersection. Every element of A is in U, because every element is in the universal set. Therefore every element of A is in $A \cap U$. This shows that $A \cap U$ and A have the same elements, so they are the same set.

$$\begin{array}{rcll} (A \cap B) \cup (A \cap \overline{B}) & = & A \cap (B \cup \overline{B}) & \text{Distributive law} \\ & = & A \cap U & \text{Because } X \cup \overline{X} = U \\ & = & A & \text{Because } X \cap U = A \end{array}$$

13. Let A be the set $\{1, 3, 5\}$ and let B be $\{2, 4, 6\}$. List the elements of the set

$$\{X \in \mathcal{P}(A \cup B) \mid X \cap A \text{ has the same number of elements as } X \cap B\}.$$

Solution.

$$\emptyset, \{1,2\}, \{1,4\}, \{1,6\}, \{3,2\}, \{3,4\}, \{3,6\}, \{5,2\}, \{5,4\}, \{5,6\}, \{1,3,2,4\}, \{1,3,2,6\}, \{1,3,4,6\},$$
$$\{1,5,2,4\}, \{1,5,2,6\}, \{1,5,4,6\}, \{3,5,2,4\}, \{3,5,2,6\}, \{3,5,4,6\}, \{1,3,5,2,4,6\}$$

14. What are the elements of $\mathcal{P}(\{\emptyset, \{\emptyset\}\})$?

Solution.

$$\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\emptyset\}\}$$

16. (*) Assume we have an algorithm which has a running time equal to $T(n)$ when the input has size n. We know that $T(1) = 3$ and

$$T(n+1) = 2T(n) - n + 1$$

when n is a positive integer.

(a) By calculating values of $T(n)$, can you guess a formula for $T(n)$?

Solution.

n	1	2	3	4	5	6
$T(n)$	3	6	11	20	37	70

This sequence looks vaguely exponential, since it almost doubles at every step. So let's compare it to the powers of 2 by calculating $T(n) - 2^n$.

n	1	2	3	4	5	6
2^n	2	4	8	16	32	64
$T(n) - 2^n$	1	2	3	4	5	6

From this table we might guess that $T(n) - 2^n = n$, so $T(n) = 2^n + n$.

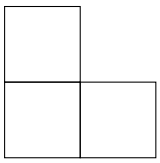
(b) Are you able to use induction to prove that your formula is correct for every value of n?

Solution. We prove by induction that $T(n) = 2^n + n$ for every positive integer n. First we check the base case, $n = 1$. Because $T(1) = 3$ and $2^1 + 1 = 3$, the base case holds. Next we make the inductive assumption that $T(k) = 2^k + k$. We will use this assumption to prove that $T(k+1) = 2^{k+1} + (k+1)$. By the definition of the T function we see that $T(k+1) = 2T(k) - k + 1$. The inductive hypothesis says that $T(k) = 2^k + k$ so

$$T(k+1) = 2T(k) - k + 1 = 2(2^k + k) - k + 1 = 2 \cdot 2^k + 2k - k + 1 = 2^{k+1} + (k+1),$$

We have shown that $T(k) = 2^k + k$ implies $T(k+1) = 2^{k+1} + (k+1)$ and this completes the proof.

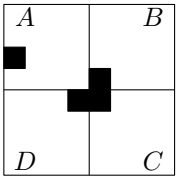
17. (*) A *tromino* is a tile made up of three squares, as in the picture below.



Let n be a positive integer. Prove by induction on n that if we take a $2^n \times 2^n$ grid of squares, and remove any one of these squares, then we can cover the remaining squares by tromino-shaped tiles in such a way that no tiles overlap. (Note that we are allowed to rotate the tiles.)

Solution. We check the base case, $n = 1$. In this case, we are removing a single square from a $2^1 \times 2^1$ grid, which is a 2×2 grid. Removing a single square from a 2×2 grid produces a tromino, so the base case is true.

Now we make the inductive assumption that the result holds when $n = k$. We will use this assumption to prove that the result holds when $n = k + 1$. So imagine we have a $2^{k+1} \times 2^{k+1}$ grid. We divide this into four quarters, labelled A, B, C, and D. Each of these quarters is a $2^k \times 2^k$ grid. Assume that the square we have removed is from quarter A. (If it is from one of the others, we can just relabel the quarters.) By the inductive assumption, the remaining squares in A can be tiled by trominoes. Now we remove one square from each of B, C, and D, chosen in such a way that the three removed squares form a tromino. Each of B, C, and D with a square removed can be tiled by trominoes, by induction, and then we fill the remaining three squares with a single tromino. So we have tiled the entire space with trominoes, and the proof is complete.



18. (*)

(a) Express 1, 3, 5, and 7 in the form $a^2 - b^2$, where a and b are non-negative integers.
Solution.

$$1 = 1^2 - 0^2 \qquad 3 = 2^2 - 1^2 \qquad 5 = 3^2 - 2^2 \qquad 7 = 4^2 - 3^2$$

(b) From this experiment, can you see a way to directly prove that every odd positive integer can be expressed as $a^2 - b^2$, where a and b are non-negative integers?
Solution. An odd positive integer can be expressed in the form $2k + 1$, where k is an integer. For example, $7 = 2 \cdot 3 + 1$. From the experiment in part (a), it appears that $2k + 1$ can be expressed in the form $(k + 1)^2 - k^2$. We will try to prove this by direct calculation.

$$(k + 1)^2 - k^2 = (k + 1)(k + 1) - k^2 = (k^2 + k + k + 1) - k^2 = 2k + 1.$$

This shows that any odd number $2k + 1$ can be expressed as $(k + 1)^2 - k^2$. So we let a be $k + 1$ and we let b be k, and the proof is complete.

19. (*) Let A be a set and let k be a positive integer. A *partition* of A into k *parts* is a collection of k non-empty subsets of A, such that every element of A is in exactly one of these subsets. For example, if A is $\{1, 2, 3, 4, 5, 6\}$, then $\{1, 3\}, \{2\}, \{4, 5, 6\}$ is a partition of A into 3 parts.

(a) Let A be $\{1, 2, 3, 4\}$. Find all partitions of A into 2 parts, and all partitions into 3 parts.
Solutions. These are the partitions into 2 parts.

$\{1\}, \{2, 3, 4\}$ $\{2\}, \{1, 3, 4\}$ $\{3\}, \{1, 2, 4\}$ $\{4\}, \{1, 2, 3\}$ $\{1, 2\}, \{3, 4\}$ $\{1, 3\}, \{2, 4\}$ $\{1, 4\}, \{2, 3\}$

These are the partitions into 3 parts.

$\{1\}, \{2\}, \{3, 4\}$ $\{1\}, \{3\}, \{2, 4\}$ $\{1\}, \{4\}, \{2, 3\}$ $\{2\}, \{3\}, \{1, 4\}$ $\{2\}, \{4\}, \{1, 3\}$ $\{3\}, \{4\}, \{1, 2\}$

(b) If n is a positive integer and k is an integer satisfying $1 \leq k \leq n$, then $S(n,k)$ stands for the numbers of partitions into k parts of a set with n elements. This is called a *Stirling number of the second kind*. Prove that $S(n+1,k) = kS(n,k) + S(n,k-1)$. (This fact was first discovered in 1782 by the Japanese mathematician Masanobu Saka.)

Solution. Let A be a set of size $n+1$. We want to know the number of partitions of A into k parts. Choose an arbitrary element a of A. The partitions of A into k come in two types: either a is in a set all by itself, or it is not. In a partition of the first type, the remaining n elements need to be partitioned into $k-1$ parts. There are $S(n,k-1)$ ways to do this. The second type of partition corresponds to the partitions of $A - \{a\}$ into k parts. We can add a to any one of the sets in such a partition, and there are k ways of doing this, as there are k parts. There are $S(n,k)$ partitions of $A - \{a\}$ into k parts, so there are $kS(n,k)$ ways of creating a partition of A into k parts where a is not in a set all by itself. In summary, there are $kS(n,k) + S(n,k-1)$ ways of partitioning A into k parts.